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Detecting Violence in Videos using Convolutional Neural Networks

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Abstract

This paper explores the practical implementation of 3D Convolutional Neural Networks (CNNs) for real-time violence detection in surveillance scenarios, focusing on applications such as traffic aggression and bullying detection. In recent years, the proliferation of surveillance cameras has provided a vast amount of visual data, necessitating efficient and accurate automated methods for threat identification. In this study, we investigate the feasibility and effectiveness of employing 3D CNNs for violence detection in real-life scenarios. We propose a comprehensive framework that integrates data augmentation techniques and fine-tuning strategies to address the challenges of limited annotated data and diverse environmental conditions. Furthermore, we conduct extensive experiments on benchmark datasets to evaluate the performance of the proposed approach in detecting physical altercations. In summary, this study highlights the efficacy of 3D CNNs in detecting physical altercations, thereby contributing to the advancement of automated surveillance systems aimed at enhancing public safety. The findings presented herein not only underscore the practical utility of deep learning techniques in addressing real-world challenges but also point towards future avenues for improving the accuracy and reliability of violence detection algorithms through multimodal integration.

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1. Introduction

From an early age, we acquire knowledge of colors through observation. We associate the sky with blue, the sun with yellow, blood with red, and the moon with white. Subsequently, we endeavor to apply this understanding to recognize these colors in various objects, solidifying our comprehension. Similarly, a CNN operates on the same principle when it comes to color recognition. Analyzing numerous images that exhibit distinct color differentiations, can

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produce remarkable outcomes. Machine learning yields models that effectively address specific tasks. These models encompass geometric, probabilistic, and logical frameworks, alongside various classification techniques. Examples of such techniques include decision trees, genetic algorithms, artificial neural networks, support vector machines, and Bayesian networks.

The model proposed in this research paper is centered around a CNN designed to identify instances of physical violence. It takes multiple videos as input, whether they *contain aggression or not*. Through training, the network acquires the ability to detect violence in videos, serving purposes such as public safety monitoring, identifying aggression on the road, and even monitoring children's behavior in educational settings to prevent bullying.

In what follows, Section 2 presents the state of the art, and Section 3 presents the details of the proposed solution. Section 4 provides an overall overview of the results obtained and the evaluations made. Finally, the conclusions of the work and possible future research directions are presented.

2. State of the Art

There are numerous studies conducted on violence detection, including the LHOG and LHOFF Case Study, CNN and LSTM Case Study, and 3D CNN.

- An example for **LHOG and LHOFF Case Study** is in [1], where optical flow field distribution is utilized for motion region segmentation. Within these regions, two types of low-level features are extracted to represent violent behaviors' appearance and dynamics. The two methods are the Local Histogram of Oriented Gradient (LHOG) from RGB images and the Local Histogram of Optical Flow (LHOFF) from optical flow images. These features are encoded using the Bag of Words (BoW) model to eliminate redundant information and generate a specific length vector for each video. Subsequently, the video vectors are classified using Support Vector Machine (SVM) [1]. The process involves video preprocessing, motion region segmentation, feature extraction, feature processing, and prediction. Similar approaches are presented in [2, 3].
- The architecture described in [4] utilizes a CNN for feature extraction, followed by a **long short-term memory network** (LSTM) for sequence prediction based on the extracted characteristic vectors. The Xception network architecture, pre-trained on the ImageNet dataset [4], was chosen for this purpose. By leveraging the pre-trained model, resources and time are saved as only the last 4 layers are retrained while the initial layers remain unchanged. The resulting model achieves a classification time of approximately 7.89 milliseconds per frame, making it well-suited for real-time violence detection. Similar approaches are presented in [5, 6].
- Another approach to identifying violence involves utilizing a **3D CNN** as outlined in reference [7]. This method is capable of recognizing intricate sequential patterns to forecast violence with a high level of precision. Initially, the camera captures the video stream, which is then inputted directly into a MobileNet CNN model that has been trained to identify individuals. Upon detecting a person in the video, a sequence of 16 frames is sent to the 3D CNN model to extract spatiotemporal characteristics. These characteristics are subsequently passed to a Softmax classifier for a comprehensive analysis and prediction. If violence is detected, an alert is dispatched to the nearest security department. Similar approaches are presented in [8, 9, 10].

3. Proposed Solution

Our objective is to identify instances of violence by utilizing a 3D CNN architecture model. This approach enables us to effectively grasp intricate sequential patterns and accurately forecast occurrences of violence. CNNs have made significant advancements, demonstrating superior accuracy and improved outcomes in a range of computer vision applications, including behavior recognition, security, object tracking, and activity recognition. The success of CNNs in these areas has paved the way for the development of a 3D CNN-based violence detection model, showcasing their potential impact and versatility. *The key contributions* in this paper are:

- A challenging problem that lends itself to the capabilities of neural networks is the identification of *complex patterns through a 3D CNN* model. This model excels at learning intricate sequences and can predict violence with *remarkable accuracy*.

- In this paper it is utilized *Grid Search* [11] as a technique to explore *the best hyperparameters*. The intention was to ensure the creation of an optimal model.
- The detection of individuals will be performed using *YOLO v8n* (nano version). We utilized the widely recognized YOLO algorithm (You Only Look Once), to identify individuals. This algorithm is renowned for its ability to swiftly process images, enabling real-time object detection. Moreover, *YOLO* is *highly regarded* due to its implementation of a *superior neural network architecture* that yields precise results [12, 13].
- To evaluate the model, we specifically selected *movie trailers*. We save and process *16 frames* from these trailers only if a person is detected with a confidence level exceeding 0.5%. The preprocessing steps applied to the initial training data are also applied to these frames. This stack of frames is then sent to the model for violence detection.

3.1. The Dataset

The violence detection model is trained using a combination of three data sets, which are not balanced as seen in their distribution in Figure 1.

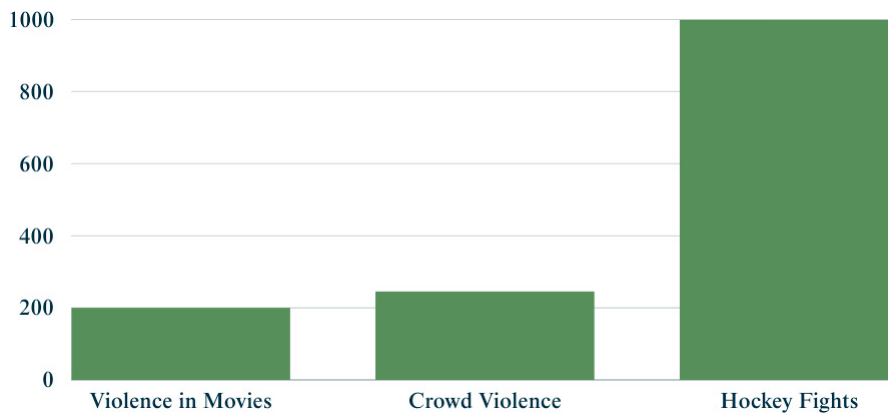


Fig. 1: The data set used in training the proposed model.

- We came across the *Violence in Movies* dataset while reading an article referenced as [7], which was introduced by Nievas. This dataset comprises 200 videos featuring violent scenes extracted from action movies. Additionally, non-violent scenes are sourced from publicly available action recognition datasets.



Fig. 2: Frames from the Violence in Movies dataset classified as non-violent.

- The *Crowd Violence* dataset is presented by Hassner in paper [14]. This dataset contains 246 YouTube videos. The original content consisted of five groups of videos, each group containing two categories: violent and non-violent.



Fig. 3: Frames from the Violence in Movies dataset classified as violent.



Fig. 4: Frames from the Crowd Violence dataset classified as non-violent.



Fig. 5: Frames from the Crowd Violence dataset classified as violent.

- We discovered the *Hockey Fights dataset* in paper [7], where it was introduced by Nievas. This dataset consists of 1,000 videos from the National Hockey League, with 500 classified as violent and the other 500 as non-violent.



Fig. 6: Frames from the Hockey Fights dataset classified as non-violent.

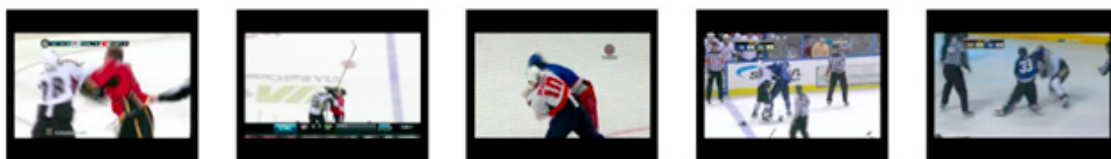


Fig. 7: Frames from the Hockey Fights dataset classified as violent.

3.2. Data Preparation

The final dataset comprises a total of 1,446 videos, which are categorized into violent and non-violent. Each category consists of 723 videos, and the total is evenly divided. We have saved the file path for these videos. The distribution for training, validation, and testing is as follows: 70%, 20%, and 10% respectively. For training the model, we utilized 70% of the dataset, which amounts to 506 videos from each category. The model was then validated using

20% of the dataset, which equates to 145 videos from each category. Finally, the model was tested using 10% of the dataset, which accounts for 72 videos from each category. To utilize this data, *the videos need to be converted into frames*.

We have employed a script that converts the videos into frames, resulting in the creation of folders named after the respective videos (e.g., *video.avi* or *video.mpg*). Each frame within these folders is numbered from 1.jpg to the maximum number of possible frames. Subsequently, *the frames are organized into stacks*. Each stack consists of 16 frames. To keep track of the video paths, the starting point of each stack, and the corresponding label (0 for non-violent and 1 for violent), We have utilized three additional files: *train.txt*, *val.txt*, and *test.txt*. This was necessary due to the overlapping sequence of frames, where 8 out of 16 frames overlap between two consecutive videos [7].

3.3. Data Preprocessing

The stack's images are initially resized to 128×171 pixels, then square-cropped to 112×112 pixels to align with the 3D CNN's input shape. Subsequently, the image is transformed into a PyTorch tensor, and each RGB frame is normalized using $mean = [0.485, 0.456, 0.406]$ and $std = [0.229, 0.224, 0.225]$. This standard transformation technique is based on the normalization method from ImageNet [15]. Figure 8 depicts the preprocessing and normalization steps. Following the creation of the stacks, they were saved as *train.h5*, *val.h5*, and *test.h5* files.

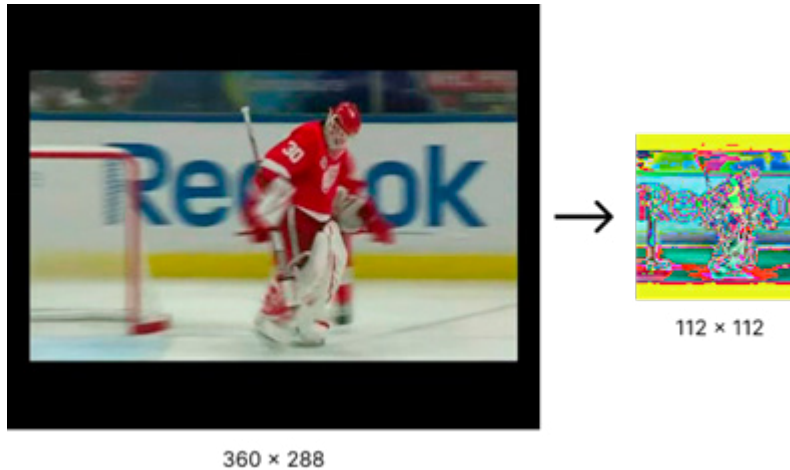


Fig. 8: Preprocessing and normalizing a frame from the dataset.

3.4. The Proposed Model

The performance of 3D CNN has served as the inspiration for the proposed model. This network is comprised of eight convolutional layers, five max-pooling layers, and two fully connected layers, culminating in a sigmoidal output layer. The convolutional layers utilize $3 \times 3 \times 3$ kernels with a stride, while the max pooling layers employ $2 \times 2 \times 2$ kernel sizes, except for the initial layer which uses $1 \times 2 \times 2$, with two strides to retain time-based information. The number of filters in each convolutional layer is 64 for the first, 128 for the second and 256 for the third layer. The fully connected layers, *fc6*, and *fc7*, consist of 4,096 neurons, and the Sigmoid layer produces two outputs: violent - 1 and non-violent - 0. The comprehensive architecture is visually depicted in Figure 9.

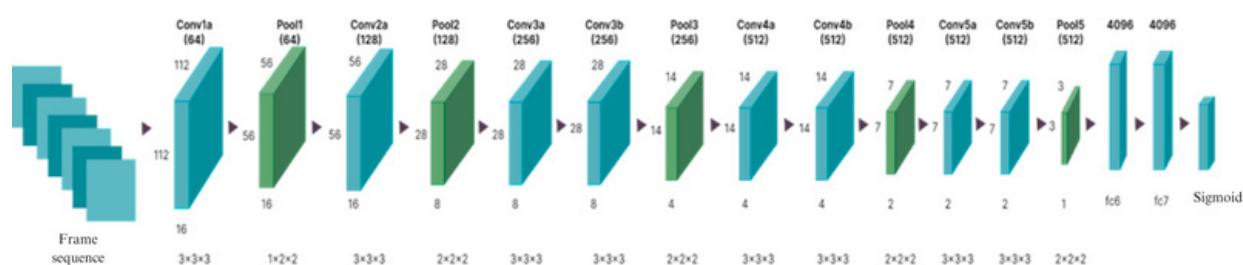


Fig. 9: The proposed 3D CNN model comprises eight convolutional layers, five max-pooling layers, and two fully connected layers, followed by the sigmoidal layer. The convolutional layers are depicted in blue, while the pooling layers are shown in green. The elongated rectangles at the end symbolize the fully connected layers, and the last small rectangle represents the sigmoidal layer.

3.5. Optimization of Hyperparameters

Grid Search [11] is a method used to find the best hyperparameter values for a model through an exhaustive search on specific parameter values. The model also referred to as an estimator, is fine-tuned by exploring various combinations of hyperparameters. To determine the optimal hyperparameters for our model, we conducted a Grid Search on the number of epochs, batch size, learning rate, and dropout layer threshold. After spending 21, 899 seconds, which is approximately 6 hours, on this search, we identified the most suitable hyperparameters: 10 epochs, 30 batch size, 0.0001 learning rate, and 0.5 dropout layer threshold.

3.6. Final Model

The ultimate design showcases a 3D CNN structure consisting of eight convolutional layers, five max-pooling layers, two fully connected layers, and one sigmoidal layer. The API hosted in an EC2 instance allows for *universal access to the model*. There are two distinct endpoints available: one for uploading videos and another for utilizing the model itself. The video upload endpoint utilizes the AWS Lambda function to preprocess the data, store the frames in the DunamoDB database, and provide the user with an identification number. Subsequently, the user can invoke the second endpoint using the provided ID to execute the model on the video. The endpoint will then respond with the percentage of violence in the video. This flow can be visualized in Figure 10.

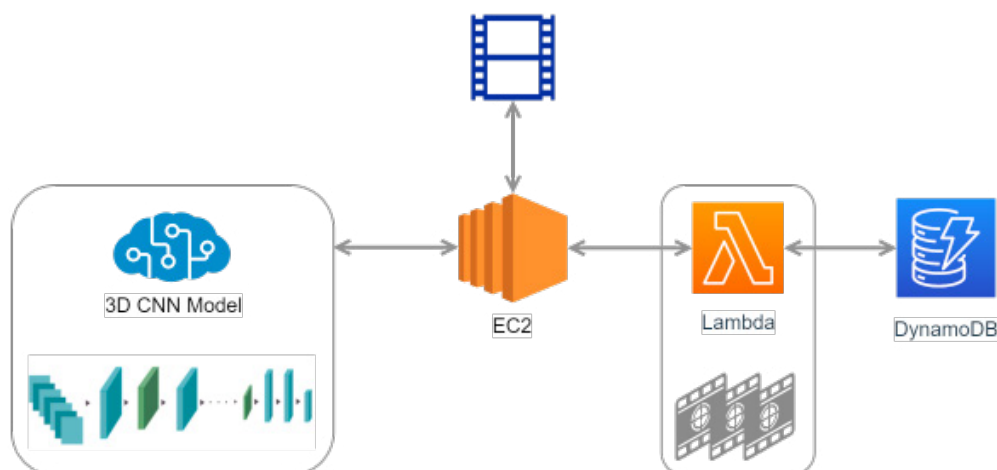


Fig. 10: Final System Architecture.

4. Results and Evaluation

4.1. Model Results

We reserved a 10% portion of the data for testing purposes, and the resulting test accuracy stands at 87.25%. To compare the outcomes of the 3D CNN model, we divided the three test datasets and assessed each one individually.

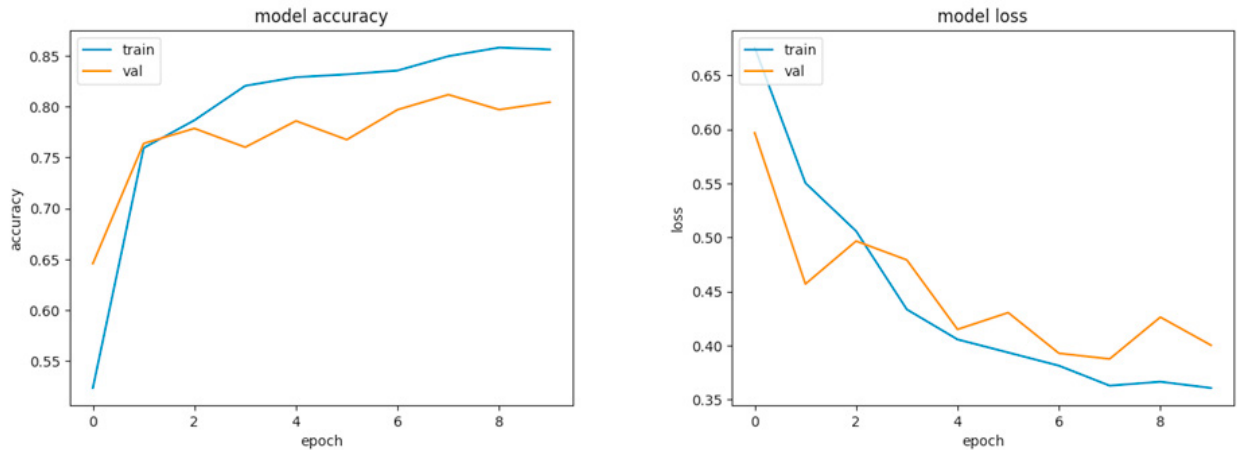


Fig. 11: Graphs showing model accuracy in left and loss in right.

Table 1 displays the obtained result and the results mentioned in reference [7] of the article.

Methods	Violence in Movies	Violent Crowd	Hockey Fights
ViF, OViF, AdaBoost and SVM	-	88.0	87.5
Hough Forest and 2D CNN	99.0	-	94.6
STIP, BoW and SVM	89.5	-	-
Motion Blobs and Random Forest	96.5	-	-
ViF	-	81.3	82.9
sHOT	-	82.2	-
Improved Fished Vectors	99.5	96.4	93.7
3D CNN article	99.9	98.0	96
Proposed solution	86.84	83.33	91.58

Table 1: Comparison with existing results, table taken and modified from reference [7].

To summarize, the developed model demonstrates a comparable accuracy level to the already established models. It is worth mentioning that the accuracy is slightly lower when dealing with crowds, as distinguishing between peaceful and violent crowds poses a challenge.

4.2. Experiments

To evaluate the trained model, we opted to conduct various experiments: assessing the model's performance on videos, evaluating the model's accuracy on trailers, and fine-tuning the model using a different data set.

4.2.1. Testing the Model on Videos

We decided to conduct a series of tests on multiple videos¹ in order *gain a deeper understanding of the factors that lead the model to classify frames as violent*. Each video was examined, and whenever individuals were detected using YOLO v8n, we saved 16 frames in a list to be utilized by the violence detection model. Once the 16 frames were saved, the initial 8 frames were removed, and the subsequent 8 frames became the next frames in the video sequence. Following a comprehensive analysis, we made the following discoveries:

- **The man assaults a child**². In this video, there is a clear depiction of a man assaulting a child. Both individuals, a man and a child, are visible in the footage. The man forcefully grabs the child's sweatshirt, constituting an act of assault. Due to the abrupt physical contact, the model is capable of identifying violence with an accuracy rate of 80%.
- **The sea**³. A woman looking at the sea is correctly classified as nonviolent.
- **Activities**⁴. Several people are being filmed, and the frames change quite quickly, but the model can classify them as non-violent. What is interesting about this video is the moment when a boy hits a ball hard with a baseball bat, as it is classified as violent. This is because the model perceives sudden movements as violent.
- **Abuse**⁵. The video depicts a man mistreating a woman. The model does not categorize the man's threats as violent when he raises his hand as if to strike her but identifies the violence when the man physically lashes out at the woman.
- **People walking**⁶. This video is categorized as violent due to the presence of numerous crowds. The model's classification of videos is not consistently accurate when there is an abundance of people standing closely together in the frame, as it fails to differentiate between aggressive crowds and peaceful gatherings.

These are merely a few of the video experiments we have conducted, yet we believe they hold significant relevance. *The purpose of these experiments was to showcase the model's capability to identify violence in diverse contexts beyond its original training.* Remarkably, the model exhibited a high level of accuracy in correctly categorizing violent scenes.

4.2.2. Model Testing on Trailers

The model was tested on movie trailers to determine *the minimum number of violent sequences required for the film to be labeled as violent*. It is important to note that each sequence consists of 16 frames. Since trailers serve as a preview of the movie, they should contain enough violent scenes to give viewers an idea of what to expect. The methodology employed was similar to the video experiment, where frames were fed into the violence detection model only when individuals were detected by the YOLO v8n model. Upon conducting a detailed analysis, the following findings were uncovered:

- **Kill Bill**⁷. In this trailer, the model detects 95 violent sequences and 212 non-violent sequences. This means that roughly 30% of the trailer consists of violent sequences.
- **Slap Shot**⁸. The model detects 114 violent sequences and 182 non-violent sequences, which means that 38% of the video consists of violent sequences.
- **The Karate Kid**⁹. In the trailer, the model detects 114 violent sequences and 228 non-violent sequences, so 33% of the content is violent.

¹ Videos can be downloaded from here: <https://www.pexels.com/ro-ro/>

² It can be viewed here: <https://youtu.be/SgNwsaj91mE>

³ It can be viewed here: <https://youtu.be/9KiPDQWxbfQ>

⁴ It can be viewed here: <https://youtu.be/CZ1QF3VbSNA>

⁵ It can be viewed here: <https://youtu.be/yxWAgm5giQ>

⁶ It can be viewed here: https://youtu.be/x_-xQab5rvM

⁷ It can be viewed here: <https://youtu.be/81z3T24Z0F8>

⁸ It can be viewed here: https://youtu.be/xBUagWZ_iro

⁹ It can be viewed here: <https://youtu.be/KuGK11NfJGg>

Upon concluding this experiment, we discovered that films containing a significant amount of violence possess trailers that exhibit *a violent content exceeding 30%*. Simultaneously, we recognized that the model exhibits the ability to identify instances of violence in diverse settings, surpassing the ones it was trained on.

4.2.3. Training the Model on a Different Dataset

We tried to train the model on the **RWF-2000** dataset. This dataset *contains 2,000 videos extracted from surveillance cameras* with real scenes from all over the world and collected from YouTube. The accuracy of the 3D CNN model on the RWF-2000 data set in reference [16] is 87%. Unfortunately, fast movements can be misinterpreted, especially if the quality of the frames is not the best.

The model could not be trained using this dataset due to insufficient computational resources, as the videos consist of an average of 17 frames, which is significantly higher compared to the 8-10 frames in the other three datasets. Additionally, the dataset contains numerous irrelevant frames that consume a large amount of memory and do not contribute to the effective training of the model.

5. Conclusion

The model created and presented in this paper is based on a 3D convolutional neural network that identifies physical violence. This model was trained on several databases. To have relevant data the videos were replaced with frames over which the necessary preprocessing was applied. The model was adjusted following a Grid Search with the most optimal hyperparameters, and the accuracy of the model is 87.25%. The model was tested on several videos, and the frames were sent to the model only if at least one person was detected with YOLO v9.

In conclusion, *the proposed violence detection model is suitable for practical use due to its high accuracy*. The process involves three steps: (1) identifying individuals with a pre-trained YOLO v8n model, (2) preprocessing frames after reaching a count of 16, and (3) utilizing a 3D CNN model for violence detection. This model's ability to recognize violence stems from its capacity to learn intricate pattern sequences. Upon analyzing predictions in the final stage, decisions are made. We are confident that this model can effectively identify and flag violence in surveillance camera footage. While the primary objective has been met, there is potential for enhancement to prevent false alarms.

5.1. Future Work

Violence detection poses a significant challenge due to its intricate nature and high computational demands. The current model we have developed can be enhanced by employing a work environment that aligns with the complexity of this issue. In the future, if computational resources become more accessible, we can create a new dataset for training the model. This dataset would consist of the *three datasets we have utilized thus far, along with a fourth dataset called RWF*. To *optimize RWF*, it is crucial to eliminate unnecessary frames where individuals are not present, thereby conserving memory. By incorporating a larger dataset, we can encompass more instances of violence, enabling the model to identify such situations with even greater precision. Additionally, *augmenting the model with sound* can further aid in the detection of violent scenarios.

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